COFACE MAPS OF A COSIMPLICIAL OBJECT ARE SPLIT MONOMORPHISMS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the identity sigma^{n-1}_i delta^n_i = id is one of the standard cosimplicial relations; the author's 'for j < n' is read as 'for i < n'.

Editorial note. The label lemma-di-injective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Lemma 0.1. *Let $\mathcal{C}$ be a category. Let U be a cosimplicial object of $\mathcal{C}$. Each of the morphisms $\delta_i^n : U_{n-1} \to U_n$ has a left inverse. In particular δ_i^n is a monomorphism.*

Proof. This is true because $\sigma_i^{n-1} \circ \delta_i^n = \mathrm{id}_{U_n}$ for $j < n$. □

Date: stacks-project:simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.